

Assignment Specification: Interactive 3D/VR Environments**Module:** Graphics and Visualization (Joint Project with Intelligent Systems)**Group Size:** 4 Students**Submission:** Source Code and a 3-Minute Demo Video**Release Date:** 31st of March 2026**Submission Deadline:** 3rd of May 2026**Total Weight:** 20% of Final Grade**1. Project Overview**

Your group is tasked with designing and developing an interactive 3D or VR simulation/game. While the core mechanic involves autonomous AI agents pathfinding through the world (evaluated in the IS module), **this module's assessment focuses entirely on the visual fidelity, 3D modeling, environment design, and physical interaction mechanics.**

2. Group Roles & Individual Responsibilities

To ensure equal contribution, your group of 4 must assign each member to one of the following specific cross-functional roles. **Your individual grade will be based on your Git commits for your specific Graphics responsibility.**

- **Student 1 (The World Builder)**
 - **Graphics Duty:** Level Design, Lighting, and Texturing. Responsible for constructing the 3D environment layout, applying cohesive textures, and baking the lighting and initial NavMesh.
- **Student 2 (The Systems Engineer)**
 - **Graphics Duty:** Player Interaction Physics. Responsible for programming the physics and scripts that allow the human player to move barricades, open doors, or throw objects.
- **Student 3 (The Core Developer)**
 - **Graphics Duty:** Custom 3D Modeling. Responsible for designing, modeling, and importing at least two custom 3D models from scratch using Blender or Maya.
- **Student 4 (The Agent Controller)**
 - **Graphics Duty:** Agent Animation & Movement. Responsible for translating the mathematical path arrays (from the IS module) into smooth 3D character movement, rotations, and animations in the engine.

3. Project Milestones & Deliverables

- **Week 11 - Concept & Scope Pitch:** A live session presenting your game concept, art direction, and declaring which student is taking which role.
- **Week 13 - Logic & Asset Prototype:** Custom 3D models drafted and imported. Player interaction mechanics (physics) functional.

- **Week 15 - Final Showcase & Viva:** A live demonstration of the completed simulation, followed by individual technical questioning.

4. Marking Rubric (100%)

Note: This rubric evaluates only the Graphics & Visualization components.

Assessment Type	Specific Criterion	Max Marks	Expectations for Full Marks
GROUP (30%)	Visual Cohesion	15	The individual assets (models, textures, lighting, animations) blend seamlessly together into a unified aesthetic, rather than looking like disconnected pieces.
	Build Stability	15	The final project compiles and runs smoothly. Individual physics and interactions do not conflict with or break the environment geometry.
INDIVIDUAL (70%)	Role-Specific Component Quality	20	Evaluated strictly on the student's assigned role: • <i>World Builder:</i> High-quality level layout, proper NavMesh baking, and excellent lighting/texturing. • <i>Systems Engineer:</i> Smooth, bug-free player physics and environmental interactions (doors, barricades). • <i>Core Developer:</i> Excellent custom 3D models built from scratch with proper topology and UV mapping. • <i>Agent Controller:</i> Smooth 3D agent translations and natural rotations/animations that perfectly follow the math path.
	Git Commits & Cleanliness	10	Student shows a consistent history of commits over the 5 weeks for their specific role. Commit messages are clear, and assets are properly organized in the engine without bloating the project size.
	Workflow & Tools	20	Can fluently explain their personal step-by-step development process and mastery of the specific tools used (e.g., Blender modeling, Unity/Unreal physics components).
	Design & Optimization Justification	20	Can articulate <i>why</i> they made certain visual or structural choices, and explain their specific optimization techniques (e.g., polygon counts, texture compression, physics fixed timesteps).
TOTAL		100	